



Higher Diploma in Computer Science

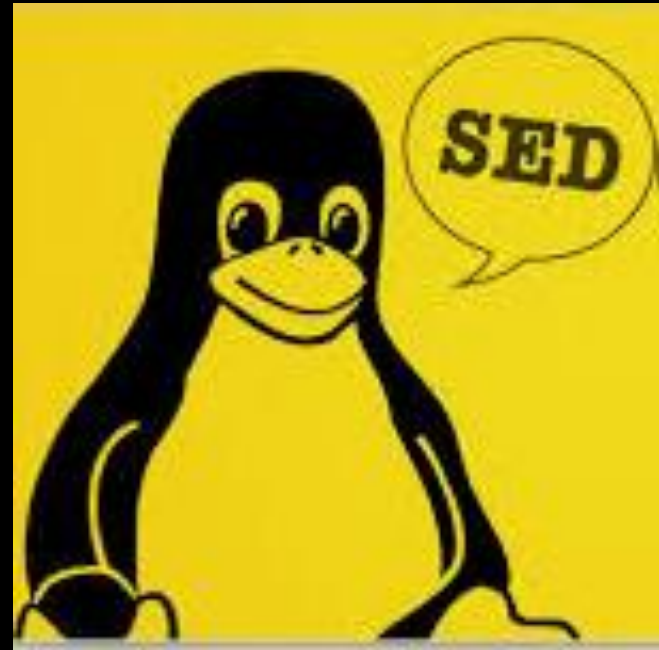
Computer Systems & Networks



Waterford Institute of Technology
INSTITIÚID TEICNEOLAÍOCHTA PHORT LAIRGE

sed

for finding, replacing,
deleting, and inserting text.



sed

- stream editor

- sed is a special editor for **modifying files** automatically
 - make global changes to a large file
 - Allows a script make changes to a file
- sed requires **an input file** and a **command** that lets sed know what actions to apply to the file
- sed can be used at the command-line, or within a shell script, to edit a file non-interactively
 - No need to open a text editor (ie to replace all occurrences of 2024 with 2025)

Find & replace

\$ sed 's/cat/dog/'

The diagram shows the command 's/cat/dog/' with labels: 's' is the command, '/' is the delimiter, 'cat' is the pattern, and 'dog' is the replacement.

\$sed -e 's/input/output/' my_file

echo every line from *my_file* to STDOUT, changing *the first occurrence* of the string 'input' on each line into 'output'.

NB sed is line-oriented, so if you wish to change *every occurrence* on each line, then you need to make it a 'greedy' global search & replace like so

\$sed -e 's/input/output/g' my_file

Syntax

- The general form is

\$sed -e '/pattern/ command' my_file

Where '*pattern*' is a regexp and

'*command*' can be one of '**s**' = search & replace, or '**p**' = print,
or '**d**' = delete, or '**i**'=insert, or '**a**'=append, etc

Redirecting output

- By default the output is written to STDOUT.
- Can redirect this to a new file:

```
sed -e 's/input/output/' my_file > new_file
```

-i Edits my_file directly, replacing all occurrences of “input” with “output”

-e is optional when running only one command: -e used to **run multiple sed commands** in a single call

```
sed -i -e 's/input/output/' my_file
```

Escape Symbol

- Edit a shell scripts to refer to **/usr/local/bin** and not **/bin** any more:

```
sed -e 's/\bin/\usr\local\bin/' my_script > new_script
```

Wildcards

- You want to take every line that starts with a number in your file and surround that number by brackets (try this with your **employee.txt** file)

```
sed -e 's/[0-9]*/(&)/' my_file
```

- Print out only directories (i.e. they begin with a “d” in ls -l)

```
$ls -l | sed -n -e '/^d/ p'
```

- You wanted to delete all lines that start with the comment symbol '#'
you could use

```
$sed -e '/^#/ d' my_file
```

- You can also use the special line number '\$' to mean 'end of file'.
 - if you wanted to delete all but the first 10 lines of a file:

```
$sed -e '11,$ d' my_file
```

- You can also use a **pattern-range** form, where the first regexp defines the start of the range, and the second the stop:

```
$sed -n -e '/caroline/,/rosanne/p' employee.txt
```

Try out this weeks Lab

03: if, while, for, break & continue; case & select for menus; using sed editor

01: Conditionals and Loops



LOOPS REPEAT ACTIONS... SO YOU DON'T HAVE TO

test · if · for · while · until

02: Loop manipulation



```
#!/bin/bash
break 10
while true; do
  echo "for, while, or until loop"
  test a fpm, until or until loop. If a is specified, break & exit after 10 loops.
  echo Status:
  for until status to 8 output & to not greater than or equal to 8
done
```

break · continue · case · select

03: sed



Stream editor · find and replace

Lab



```
#!/bin/bash
```

Loops · Menus · sed

Computer Systems > 03: Loops, Menus > Week 3 Class 1... > Lab

Advancing Scripting Skills

- 01. Conditionals
- 02. Loops
- 03. Menu driven scripts
- 04. sed**
- 05. EXER Add new records
- 06. Cisco NetAcad

04. sed

sed editor · find and replace

For this lab, download [Distros.txt file](#), a space separated text file, and try the following commands:

- Print lines 7-9 inclusive:

```
$ sed -n '7,9p' distros.txt
```

- Search for a pattern and print it:

```
$ sed -n '/SUSE/p' distros.txt # print lines matched by "SUSE"
```

- Negate the above search pattern:

```
$ sed -n '/SUSE/lp' distros.txt #except the ones matched by X
```

PRACTICE using sed

Make a copy of the **distros.txt** file, named **tempdistros.txt** and practice deleting sections of the file

```
$ sed '1d' tempdistros.txt #deletes 1st line
```

Try out this weeks Lab

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Advancing Scripting Skills

01. Conditionals

02. Loops

03. Menu driven scripts

04. sed

05. EXER Add new records

06. Cisco NetAcad

05. EXER Add new records

EXERCISE

Write a script that can be used to add a new record (employees details) to your **employee.txt** file

TIPS

Think about this logically:

- 1st check if the file exists and maybe do you want to create it if it's not there? Do you want to use to receive error messages Customised feedback?
- Then you want to give appropriate variable names in order to the positional parameter, the 1st parameter being the employee's ID, the second their name, the third their position in the company. 4th the dept they belong to and finally, 5th the expenses they claimed.
- Whats next? Time to take the details in and append them to your **employee.txt** file

ERROR CHECKING:

Error checking is an **absolute MUST** especially when you're taking user input into a system to ensure that the expected input is in the expected format i.e. email addresses, phone numbers, postcodes etc. are all in the expected format

Advancing Scripting Skills

01. Conditionals

02. Loops

03. Menu driven scripts

04. sed

05. EXER Add new records

06. Cisco NetAcad

06. Cisco NetAcad

CA · Chapter Exams

For the purpose of this module, we will today complete the final two chapters and their associated Chapter Exams.

- For the 5% CA, you are required to have completed the chapter exams from Chapter 1 up to and including Chapter 11



5% Cisco
NetAcad
chapter exams

Final NetAcad 5% CA

Completed Chapter
Exams *up to and
including* Chapter
11

